# Myanmar rCARI Validation Study Summary Report

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# 1 Introduction

# 2 Data

The data for this study was collected from June 7 to June 24, 2023, encompassing 21 camps located in the Kachin and Shan regions. To ensure a representative sample, a pre-collected household list was utilized, from which 1665 households were randomly selected. These selected households were then assigned to two distinct survey modalities: Face-to-Face (F2F) and Remote. Additionally, two different form lengths, namely Short and Long, were randomly assigned to the households. Overall, we completed a total of 1093 completed surveys for the analysis.

Table 1: Sample Completion: Face-to-Face vs. Remote

| | Not Completed | | Completed | | Total | |
|---|---|---|---|---|---|---|
| | Count | Percent | Count | Percent | Count | Percent |
| Face-to-Face | 223 | 26.77 | 610 | 73.23 | 833 | 100.00 |
| Remote | 349 | 41.95 | 483 | 58.05 | 832 | 100.00 |
| Total | 572 | 34.35 | 1093 | 65.65 | 1665 | 100.00 |

Table 1 provides an overview of the sample completion rates for the F2F and Remote survey modalities. For F2F survey, a total of 833 households were included in the sample. Out of these, 610 households successfully completed the survey, accounting for 73.23% of the F2F sample. On the other hand, 223 households did not complete the survey, representing 26.77% of the sample. For Remote, 832 households were included in the sample. Of these, 483 households completed the survey, which corresponds to a completion rate of 58.05% for the Remote sample. The remaining 349 households (41.95% of the Remote sample) did not complete the survey due to non-response and availability.

# 3 Randomization Quality

To assess the balance of the treatment (Remote) and control (F2F) groups, we examined various respondent and household characteristics, including basic demographics, income sources, and changes over time. This allowed us to evaluate whether the two groups were similar in terms of these factors, ensuring a fair comparison between the treatment and control conditions.

## 3.1 Modality Randomization

Table 2: Balance Table on Randomized Survey Modality Assignment: Face-to-Face vs. Remote

| Variable | (1)<br>F2F<br>Mean/(SE) | (2)<br>Remote<br>Mean/(SE) | (1)-(2)<br>Pairwise t-test<br>Mean difference |
|---|---|---|---|
| Respondent Age | 42.24<br>(0.54) | 38.26<br>(0.62) | 3.98*** |
| Female Respondent | 0.81<br>(0.02) | 0.80<br>(0.02) | 0.01 |
| Head of Household Respondent | 0.42<br>(0.02) | 0.35<br>(0.02) | 0.07** |
| Household Size | 5.52<br>(0.09) | 5.49<br>(0.10) | 0.03 |
| Number of household members contributing to income | 1.41<br>(0.03) | 1.41<br>(0.04) | 0.00 |
| Number of household income sources | 1.18<br>(0.03) | 1.24<br>(0.03) | -0.07* |
| Number of observations | 610 | 483 | 1093 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

## 3.2 F2F Length Randomization

Table 3: Balance Table on Randomized Survey Length Assignment in F2F Survey: Long vs. Short

| Variable | N | (1) F2F Mean/(SE) | N | (2) Remote Mean/(SE) | N | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|---|---|---|
| Respondent Age | 305 | 41.33 | 305 | 43.15 | 610 | -1.81* |
| | | (0.77) | | (0.76) | | |
| Female Respondent | 305 | 0.82 | 305 | 0.80 | 610 | 0.02 |
| | | (0.02) | | (0.02) | | |
| Head of Household Respondent | 305 | 0.43 | 305 | 0.42 | 610 | 0.01 |
| | | (0.03) | | (0.03) | | |
| Single Respondent | 305 | 0.11 | 305 | 0.07 | 610 | 0.05** |
| | | (0.02) | | (0.01) | | |
| Married Respondent | 305 | 0.66 | 305 | 0.70 | 610 | -0.05 |
| | | (0.03) | | (0.03) | | |
| IDP Household | 305 | 0.77 | 305 | 0.72 | 610 | 0.05 |
| | | (0.02) | | (0.03) | | |
| Household has disabled member | 305 | 0.17 | 305 | 0.20 | 610 | -0.03 |
| | | (0.02) | | (0.02) | | |
| Number of disabled household members | 305 | 0.18 | 305 | 0.23 | 610 | -0.05 |
| | | (0.02) | | (0.03) | | |
| Head of Household Sex | 303 | 0.60 | 304 | 0.68 | 607 | -0.08** |
| | | (0.03) | | (0.03) | | |
| Head of Household Age | 303 | 49.05 | 304 | 49.88 | 607 | -0.83 |
| | | (0.76) | | (0.74) | | |
| Household head has attended school | 303 | 0.76 | 304 | 0.82 | 607 | -0.06* |
| | | (0.02) | | (0.02) | | |
| Household Size | 305 | 5.50 | 305 | 5.54 | 610 | -0.04 |
| | | (0.13) | | (0.13) | | |
| Total number of male household member | 305 | 2.73 | 305 | 2.62 | 610 | 0.11 |
| | | (0.09) | | (0.08) | | |
| Total number of female household member | 305 | 2.76 | 305 | 2.91 | 610 | -0.15 |
| | | (0.08) | | (0.09) | | |
| Total number of children | 305 | 2.17 | 305 | 2.24 | 610 | -0.07 |
| | | (0.08) | | (0.09) | | |
| Total number of senior | 305 | 0.46 | 305 | 0.48 | 610 | -0.02 |
| | | (0.04) | | (0.04) | | |
| Total number of adults | 305 | 2.86 | 305 | 2.82 | 610 | 0.05 |
| | | (0.09) | | (0.09) | | |
| Household total number of dependents | 305 | 2.63 | 305 | 2.72 | 610 | -0.09 |
| | | (0.09) | | (0.09) | | |
| Household Female/Male Ratio | 305 | 1.69 | 305 | 1.63 | 610 | 0.06 |
| | | (0.11) | | (0.10) | | |
| Household Dependency Ratio | 305 | 1.22 | 305 | 1.32 | 610 | -0.10 |
| | | (0.06) | | (0.07) | | |
| Number of household members contributing to income | 305 | 1.49 | 305 | 1.34 | 610 | 0.15** |
| | | (0.05) | | (0.04) | | |
| Number of household income sources | 305 | 1.21 | 305 | 1.15 | 610 | 0.06 |
| | | (0.04) | | (0.03) | | |
| Income Source: None | 305 | 0.11 | 305 | 0.09 | 610 | 0.02 |
| | | (0.02) | | (0.02) | | |
| Income Source: Skilled Waged Labor | 305 | 0.12 | 305 | 0.12 | 610 | -0.01 |
| | | (0.02) | | (0.02) | | |
| Income Source: Unskilled Waged Labor | 305 | 0.49 | 305 | 0.50 | 610 | -0.01 |
| | | (0.03) | | (0.03) | | |
| Income Source: Trade | 305 | 0.10 | 305 | 0.06 | 610 | 0.04 |
| | | (0.02) | | (0.01) | | |
| Income Source: Agricultural Production | 305 | 0.11 | 305 | 0.13 | 610 | -0.01 |
| | | (0.02) | | (0.02) | | |
| Income Change: None | 305 | 0.40 | 305 | 0.44 | 610 | -0.03 |
| | | (0.03) | | (0.03) | | |
| Income Change: Increase | 305 | 0.12 | 305 | 0.11 | 610 | 0.02 |
| | | (0.02) | | (0.02) | | |
| Income Change: Reduced < 25% | 305 | 0.25 | 305 | 0.20 | 610 | 0.05 |
| | | (0.02) | | (0.02) | | |
| Income Change: Reduced 25% - 50% | 305 | 0.17 | 305 | 0.20 | 610 | -0.03 |
| | | (0.02) | | (0.02) | | |

## 3.3 Remote Length Randomization

Table 4: Balance Table on Randomized Survey Length Assignment in Remote Survey: Long vs. Short

| Variable | (1) F2F Mean/(SE) | (2) Remote Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| Respondent Age | 37.41 | 39.05 | -1.64 |
| | (0.83) | (0.91) | |
| Female Respondent | 0.82 | 0.78 | 0.03 |
| | (0.03) | (0.03) | |
| Head of Household Respondent | 0.35 | 0.35 | -0.00 |
| | (0.03) | (0.03) | |
| Household Size | 5.58 | 5.41 | 0.17 |
| | (0.15) | (0.13) | |
| Number of household members contributing to income | 1.41 | 1.41 | 0.00 |
| | (0.05) | (0.06) | |
| Number of household income sources | 1.26 | 1.23 | 0.03 |
| | (0.04) | (0.04) | |
| Income Source: None | 0.03 | 0.06 | -0.03* |
| | (0.01) | (0.01) | |
| Income Source: Skilled Waged Labor | 0.06 | 0.09 | -0.02 |
| | (0.02) | (0.02) | |
| Income Source: Unskilled Waged Labor | 0.60 | 0.54 | 0.06 |
| | (0.03) | (0.03) | |
| Income Source: Trade | 0.03 | 0.04 | -0.00 |
| | (0.01) | (0.01) | |
| Income Source: Agricultural Production | 0.18 | 0.17 | 0.01 |
| | (0.03) | (0.02) | |
| Income Change: None | 0.39 | 0.43 | -0.04 |
| | (0.03) | (0.03) | |
| Income Change: Increase | 0.15 | 0.14 | 0.01 |
| | (0.02) | (0.02) | |
| Income Change: Reduced < 25% | 0.20 | 0.17 | 0.03 |
| | (0.03) | (0.02) | |
| Income Change: Reduced 25% - 50% | 0.20 | 0.15 | 0.05 |
| | (0.03) | (0.02) | |
| Income Change: Reduced 50+% | 0.06 | 0.11 | -0.05** |
| | (0.02) | (0.02) | |
| Number of observations | 233 | 250 | 483 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

# 4 Primary Results

## 4.1 Impact of Survey Modality on Key Food Security Outcomes

Table 5: Impact of Survey Modality: Face-to-Face vs. Remote

| Variable | (1) F2F Mean/(SE) | (2) Remote Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| Food Consumption Score | 44.32 | 43.75 | 0.57 |
| | (0.50) | (0.55) | |
| Reduced Consumption Strategies Index (rCSI) | 5.43 | 9.24 | -3.81*** |
| | (0.30) | (0.40) | |
| Maximum Livelihood Coping: None | 0.26 | 0.12 | 0.14*** |
| | (0.02) | (0.01) | |
| Maximum Livelihood Coping: Stress level | 0.46 | 0.41 | 0.05 |
| | (0.02) | (0.02) | |
| Maximum Livelihood Coping: Crisis level | 0.25 | 0.33 | -0.08*** |
| | (0.02) | (0.02) | |
| Maximum Livelihood Coping: Emergency level | 0.03 | 0.14 | -0.10*** |
| | (0.01) | (0.02) | |
| Monthly total household expenditure | 435,067.62 | 417,660.68 | 17,406.94 |
| | (9,727.81) | (8,977.45) | |
| Monthly total household expenditure per capita | 85,223.46 | 84,800.21 | 423.25 |
| | (1,888.13) | (2,094.78) | |
| Food Expenditure Share | 0.66 | 0.66 | 0.01 |
| | (0.01) | (0.01) | |
| Number of observations | 610 | 483 | 1093 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 6: Impact of Survey Length on LCS in F2F Survey: EN vs. FS

| Variable | (1) Long(EN) Mean/(SE) | (2) Short(FS) Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| Maximum Livelihood Coping: None | 0.25 | 0.26 | -0.01 |
| | (0.02) | (0.03) | |
| Maximum Livelihood Coping: Stress level | 0.47 | 0.45 | 0.02 |
| | (0.03) | (0.03) | |
| Maximum Livelihood Coping: Crisis level | 0.26 | 0.24 | 0.01 |
| | (0.03) | (0.02) | |
| Maximum Livelihood Coping: Emergency level | 0.02 | 0.05 | -0.02 |
| | (0.01) | (0.01) | |
| Number of observations | 305 | 305 | 610 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 7: Impact of Survey Modality on Income and FES : Face-to-Face vs. Remote

| Variable | (1) F2F Mean/(SE) | (2) Remote Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| Income: Regular and Unchanged/Increasing | 0.22 | 0.19 | 0.03 |
| | (0.02) | (0.02) | |
| Income: Regular and Reduced OR Irregular and Unchanged/Increasing | 0.39 | 0.41 | -0.03 |
| | (0.02) | (0.02) | |
| Income: Irregular and Reduced | 0.28 | 0.33 | -0.04 |
| | (0.02) | (0.02) | |
| Income: No income sources | 0.11 | 0.07 | 0.05*** |
| | (0.01) | (0.01) | |
| FES: <50% | 0.13 | 0.16 | -0.03 |
| | (0.01) | (0.02) | |
| FES: 50-65% | 0.30 | 0.33 | -0.03 |
| | (0.02) | (0.02) | |
| FES: 65-75% | 0.27 | 0.19 | 0.08*** |
| | (0.02) | (0.02) | |
| FES: >= 75% | 0.30 | 0.32 | -0.02 |
| | (0.02) | (0.02) | |
| Number of observations | 610 | 483 | 1093 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 8: Impact of Survey Modality on CARI and rCARI : Face-to-Face vs. Remote

| Variable | (1) F2F Mean/(SE) | (2) Remote Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| CARI: Food Secure | 0.08 | 0.04 | 0.04*** |
| | (0.01) | (0.01) | |
| CARI: Marginally Food Secure | 0.44 | 0.37 | 0.08*** |
| | (0.02) | (0.02) | |
| CARI: Moderately Food Insecure | 0.44 | 0.54 | -0.10*** |
| | (0.02) | (0.02) | |
| CARI: Severely Food Insecure | 0.03 | 0.05 | -0.02* |
| | (0.01) | (0.01) | |
| CARI: Food Insecure | 0.47 | 0.59 | -0.12*** |
| | (0.02) | (0.02) | |
| rCARI: Food Secure | 0.16 | 0.07 | 0.09*** |
| | (0.01) | (0.01) | |
| rCARI: Marginally Food Secure | 0.43 | 0.40 | 0.03 |
| | (0.02) | (0.02) | |
| rCARI: Moderately Food Insecure | 0.39 | 0.50 | -0.11*** |
| | (0.02) | (0.02) | |
| rCARI: Severely Food Insecure | 0.02 | 0.04 | -0.01 |
| | (0.01) | (0.01) | |
| rCARI: Food Insecure | 0.41 | 0.54 | -0.13*** |
| | (0.02) | (0.02) | |
| Number of observations | 610 | 483 | 1093 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 9: Impact of Survey Modality on Real CARI and rCARI : Face-to-Face vs. Remote

| Variable | (1) F2F Mean/(SE) | (2) Remote Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| CARI vs. rCARI: Food Secure | 0.08 | 0.07 | 0.02 |
| | (0.01) | (0.01) | |
| CARI vs. rCARI: Marginally Food Secure | 0.44 | 0.40 | 0.05 |
| | (0.02) | (0.02) | |
| CARI vs. rCARI: Moderately Food Insecure | 0.44 | 0.50 | -0.06* |
| | (0.02) | (0.02) | |
| CARI vs. rCARI: Severely Food Insecure | 0.03 | 0.04 | -0.00 |
| | (0.01) | (0.01) | |
| CARI vs. rCARI: Food Insecure | 0.47 | 0.54 | -0.06** |
| | (0.02) | (0.02) | |
| Number of observations | 610 | 483 | 1093 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 10: Impact of Survey Length on Expenditure in F2F Survey: Long vs. Short

| Variable | (1) Long Mean/(SE) | (2) Short Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| Total monthly food expenditure (average) | 300,045.50 | 255,380.15 | 44,665.35*** |
| | (8,309.95) | (8,202.09) | |
| Total monthly non-food expenditure (average) | 173,898.33 | 140,811.26 | 33,087.08*** |
| | (9,319.11) | (7,313.85) | |
| Monthly total household expenditure | 473,943.83 | 396,191.41 | 77,752.42*** |
| | (14,332.85) | (12,797.28) | |
| Monthly total household expenditure per capita | 93,346.30 | 77,100.63 | 16,245.67*** |
| | (2,940.57) | (2,280.92) | |
| Food Expenditure Share | 0.66 | 0.66 | -0.00 |
| | (0.01) | (0.01) | |
| Number of observations | 305 | 305 | 610 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 11: Impact of Survey Length on Expenditure in Remote Survey: Long vs. Short

| Variable | (1) Long Mean/(SE) | (2) Short Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| Total monthly food expenditure (average) | 280,289.58 (8,617.95) | 253,623.94 (8,379.99) | 26,665.63** |
| Total monthly non-food expenditure (average) | 124,607.94 (6,823.42) | 175,932.00 (6,911.94) | -51,324.06*** |
| Monthly total household expenditure | 404,897.52 (12,570.34) | 429,555.94 (12,767.94) | -24,658.43 |
| Monthly total household expenditure per capita | 82,083.90 (2,996.40) | 87,331.81 (2,925.90) | -5,247.91 |
| Food Expenditure Share | 0.71 (0.01) | 0.61 (0.01) | 0.10*** |
| Number of observations | 233 | 250 | 483 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

# 5 Gendered Difference on MDDI

MDDI questions were asked at individual level.

Figure 1: Gendered Difference in Minimal Dietary Diversity Food Groups

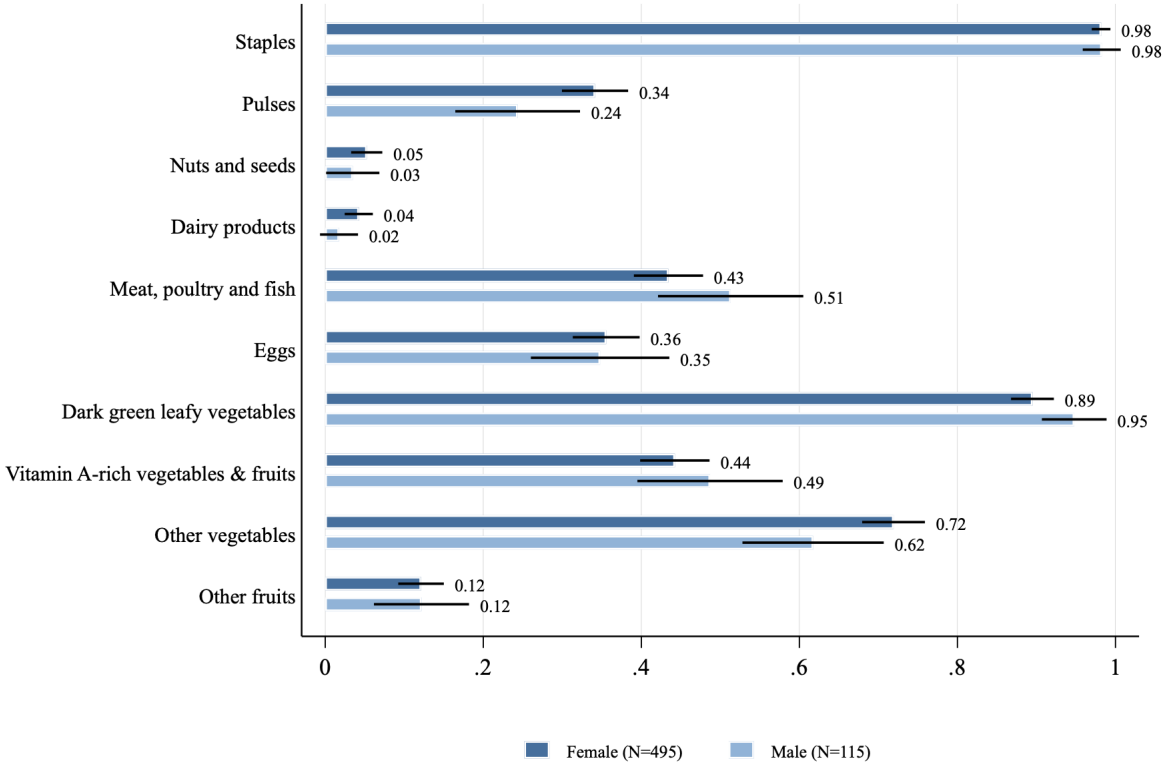

Table 12: Gendered Difference in Minimal Dietary Diversity Food Groups and Index

| Variable | (1) Female Mean/(SE) | (2) Male Mean/(SE) | (1)-(2) Pairwise t-test Mean difference |
|---|---|---|---|
| Staples | 0.98 | 0.98 | -0.00 |
| | (0.01) | (0.01) | |
| Pulses | 0.34 | 0.24 | 0.10** |
| | (0.02) | (0.04) | |
| Nuts and seeds | 0.05 | 0.03 | 0.02 |
| | (0.01) | (0.02) | |
| Dairy products | 0.04 | 0.02 | 0.03 |
| | (0.01) | (0.01) | |
| Meat, poultry and fish | 0.43 | 0.51 | -0.08 |
| | (0.02) | (0.05) | |
| Eggs | 0.36 | 0.35 | 0.01 |
| | (0.02) | (0.04) | |
| Dark green leafy vegetables | 0.89 | 0.95 | -0.05* |
| | (0.01) | (0.02) | |
| Vitamin A-rich vegetables & fruits | 0.44 | 0.49 | -0.04 |
| | (0.02) | (0.05) | |
| Other vegetables | 0.72 | 0.62 | 0.10** |
| | (0.02) | (0.05) | |
| Other fruits | 0.12 | 0.12 | -0.00 |
| | (0.01) | (0.03) | |
| MDD Index | 4.39 | 4.31 | 0.07 |
| | (0.07) | (0.13) | |
| MDD Index >= 5 | 0.44 | 0.43 | 0.01 |
| | (0.02) | (0.05) | |
| Number of observations | 495 | 115 | 610 |

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.